

ENHANCING PATHOLOGICAL SPEECH THROUGH ARTICULATORY BOTTLENECKS

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ABSTRACT

Dysarthric speech reconstruction (DSR) typically relies on linguistic or phonetic representations extracted from impaired speech to generate a more intelligible waveform. We investigate a complementary approach that instead intervenes in a representation related to speech production. Starting from a neural analysis–synthesis framework, we introduce a residual mapper that modifies an articulatory-aligned latent space while preserving speaker and prosodic information. The mapper is pretrained on parallel synthetic healthy and artificially dysarthric speech, then adapted to natural dysarthric speech using phoneme-guided and adversarial objectives. We further compare the articulatory bottleneck with a dimension-matched unsupervised representation to assess the benefit of explicit articulatory supervision. Objective and perceptual evaluations show that the proposed approach successfully improves artificially generated dysarthric speech. While the transfer to natural pathological speech remains limited, the articulatory bottleneck introduces an interpretable representation, offering a structured space for targeted speech corrections.

Index Terms— dysarthria, speech pathology, speech enhancement, speech production

1. INTRODUCTION

Dysarthria is a motor speech disorder in which impaired control of the speech musculature affects respiration, phonation, resonance, articulation, and prosody, often resulting in reduced speech intelligibility [1, 2]. Dysarthric speech reconstruction (DSR) aims to transform impaired speech into a more intelligible form while preserving the speaker’s identity [3]. Recent DSR systems [4, 5, 6] typically combine self-supervised learning (SSL) representations of dysarthric speech with explicit speaker information and neural speech generation. While these approaches have shown promising improvements, reconstructing intelligible speech, particularly for severe dysarthria, remains challenging. These systems largely approach DSR as a speech reconstruction problem, in which linguistic or phonetic content is recovered from the impaired signal and used, together with speaker information, to generate a new waveform. Here, we explore a complementary perspective: rather than reconstructing speech primarily from its recovered content, we seek to correct the impaired signal within a representation related to the speech production process. We focus in particular on articulatory movements, a natural intervention space for dysarthria, where impaired motor control can result in inaccurate or poorly coordinated movements of the speech articulators [1, 2].

Our approach builds on two lines of research. First, probing studies have shown that intermediate representations of SSL speech models encode substantial information about speech production and can be strongly correlated with articulatory trajectories [7, 8]. Second, neural speech analysis–synthesis models have exploited such representations to introduce an explicit articulatory bottleneck

between speech encoding and waveform reconstruction [9, 10, 11]. Following this approach, we consider an analysis–synthesis framework based on intermediate WavLM representations selected for their strong correspondence with articulatory data, combined with separate prosody and speaker representations and a HiFi-GAN neural vocoder for waveform reconstruction [11, 12].

We investigate two alternative bottlenecks derived from the same WavLM representations: a supervised articulatory bottleneck trained to predict articulatory trajectories, and a dimension-matched unsupervised bottleneck obtained using a learned dimensionality-reduction projection. In both cases, a lightweight residual mapper modifies only this compact representation, while the speaker and prosodic representations are preserved. The mapper is first pretrained using parallel synthetic healthy and dysarthric speech and then fine-tuned on natural dysarthric speech using a phoneme-guided CTC objective combined with adversarial regularization toward representations observed in healthy speech.

Our main contributions are: (1) Formulating dysarthric speech correction as residual mapping within a compact speech-production-related representation while preserving encoded speaker and prosodic information; (2) Introducing a two-stage training strategy combining supervised mapping from parallel synthetic speech with phoneme-guided and adversarial fine-tuning on natural dysarthric speech; (3) Comparing supervised articulatory and dimension-matched unsupervised bottlenecks derived from the same SSL representations to investigate the benefit of explicit articulatory supervision.

2. RELATED WORK

Early neural-based DSR systems follow voice conversion paradigms, learning direct mappings between dysarthric and healthy speech representations, such as vocoded features or phonetic posteriorgrams [13, 14, 15]. End-to-end encoder–decoder architectures decomposed DSR into sub-modules dedicated to linguistic content extraction, speaker identity preservation, and speech generation, often employing auxiliary objectives such as cross-modal knowledge distillation [16], adversarial speaker adaptation [17], and joint speech reconstruction and phoneme recognition objectives [18, 19, 20]. More recently, SSL-based DSR has emerged as a dominant paradigm, leveraging speech representations extracted from SSL encoders together with neural speech generators, as in Unit-DSR [4], CoLM-DSR [5], and DiffDSR [6]. Despite their architectural differences, these approaches share a common emphasis on recovering linguistic or phonetic information from dysarthric speech to guide waveform reconstruction. In parallel, neural speech analysis–synthesis systems have introduced explicit articulatory bottlenecks inferred from acoustic speech representations [9, 10, 11]. Articulatory features have also been exploited for disordered-speech ASR, including through cross-domain acoustic-to-articulatory inversion [21, 22] and data augmentation with articulatory-based synthesis [23, 24]. However, these approaches use articulation either for speech analysis–

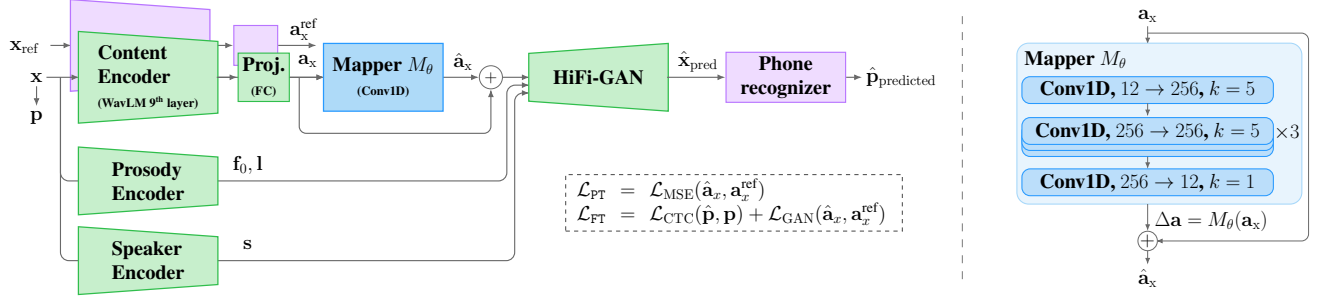


Fig. 1. Left: Proposed articulatory correction framework. Frozen encoders extract the articulatory bottleneck trajectory $\mathbf{a}_x$, prosodic features (pitch f_0 and loudness l), and speaker embedding $\mathbf{s}$ from the input waveform $\mathbf{x}$. A trainable residual Conv1D mapper predicts a correction to $\mathbf{a}_x$, yielding $\hat{\mathbf{a}}_x$, which is resynthesized by a frozen HiFi-GAN vocoder while preserving the original speaker and prosodic features. Purple boxes are used for training only. A reference audio $\mathbf{x}_{ref}$ is sent to the same articulatory bottleneck to pretrain the mapper with MSE loss. A frozen phonetic decoder provides phonetic CTC-based supervision against the target sequence $\mathbf{p}$. Fine-tuning the mapper combines CTC and adversarial losses. **Right:** Architecture of the mapper.

synthesis of typical speech or to help recognition of disordered speech, rather than as an intervention space for DSR. To the best of our knowledge, this work is the first to investigate dysarthric speech reconstruction by explicitly modifying an articulatory representation within a neural speech analysis–synthesis framework.

3. METHODOLOGY

3.1. Analysis-synthesis pipeline (identity)

Given an input dysarthric waveform $\mathbf{x}$ and its target phoneme sequence $\mathbf{p}$, our objective is to generate a waveform $\hat{\mathbf{x}}$ with improved phonetic intelligibility while preserving the speaker identity and prosodic characteristics¹ of the original signal.

Supervised articulatory bottleneck (SAB): The first proposed model is built on top of SPARC pipeline [11] represented in green in Fig. 1. A content encoder uses the ninth hidden layer of a pre-trained WavLM model [25] followed by a projection layer to get a 12-dimensional articulatory bottleneck (AB) $\mathbf{a}_x$. In SPARC, $\mathbf{a}_x$ is supervised (SAB) with an MSE loss to encode the estimated horizontal and vertical trajectories of six EMA sensors attached to the articulators (tongue, lips, jaw, see [26]) of a native speaker of English. In parallel, a frozen prosody encoder extracts the fundamental frequency f_0 and loudness l , and speaker encoder extracts an embedding $\mathbf{s}$ from the input waveform $\mathbf{x}$. A frozen HiFi-GAN-based decoder reconstructs the waveform from these representations. When the AB remains unchanged, this reconstruction is referred to as the SAB *identity* condition.

Unsupervised articulatory bottleneck (UAB): To assess whether the correction performance depends on the pseudo-EMA nature of the SAB, we construct a dimension-matched unsupervised articulatory bottleneck (UAB). The UAB model shares exactly the same encoder and decoder as the SAB model (in green in Fig. 1), matching its representation dimensionality, while replacing the supervised AB with an unsupervised latent code, using a projection layer consisting of a three-layer MLP with dimensions $1024 \rightarrow 512 \rightarrow 512 \rightarrow 12$ and GELU activations. As for SAB, the projection is applied to the ninth layer of a pre-trained WavLM, which linear probing experiments in [11] showed to be strongly correlated with EMA trajectories of a target speaker ($r = 0.878 \pm 0.012$). Thus, while UAB is not explicitly supervised to predict articulatory movements, its input representation already carries substantial information about speech production.

3.2. Articulatory mapper

The mapper M_θ shown in Fig. 1 (right) predicts a residual correction from the original bottleneck: $\hat{\mathbf{a}}_x = \mathbf{a}_x + M_\theta(\mathbf{a}_x)$, to encourage the mapper to preserve the original bottleneck when no correction is required.² The mapper consists of one-dimensional convolution with 256 channels, followed by three one-dimensional convolutional blocks with 256 channels and a final one-dimensional convolution back to the original 12-dimensional bottleneck. All intermediate convolutions use a kernel size of 5.

3.3. Training objectives

Stage 0, identity models: The SAB *identity* model uses the weights provided by SPARC³ [11]. The UAB *identity* model is constructed by jointly pretraining the projection with the HiFi-GAN decoder, using the same frozen content, prosody and speaker encoder as in SPARC to provide the input, and the same waveform reconstruction objectives and discriminators as HiFi-GAN training [12]. In the following, and for both SAB and UAB models, all encoders, projection and decoder (green in Fig. 1) remain frozen throughout training stages 1 and 2 of the mapper, described below.

Stage 1, mapper pre-training: We use parallel signals of clean and dysarthric synthetic speech ($\mathbf{x}^{ref}$ and $\mathbf{x}$) (detailed in sec. 4.1) and get their respective AB ($\mathbf{a}_x^{ref}$ and $\mathbf{a}_x$) using a shared frozen content encoder and projection. The mapper is pretrained in a supervised manner so that its output matches $\mathbf{a}_x^{ref}$: $\mathcal{L}_{PT} = \text{MSE}(\mathbf{a}_x^{ref}, \hat{\mathbf{a}}_x)$. This pre-training exposes the mapper to a wide range of controlled articulatory perturbations, while paired reference trajectories provide direct supervision toward their healthy counterparts.

Stage 2, mapper fine-tuning: The synthesized waveform $\hat{\mathbf{x}}$ is passed through a frozen phone recognizer based on Wav2Vec 2.0⁴ R , producing the predicted phoneme sequence $\hat{\mathbf{p}} = R(\hat{\mathbf{x}})$. Given the target phoneme sequence $\mathbf{p}$, the mapper is trained using the combined objective: $\mathcal{L}_{FT} = \text{CTC}(R(\hat{\mathbf{x}}), \mathbf{p}) + \lambda_{GAN} \mathcal{L}_{GAN}(\hat{\mathbf{a}}_x, \mathbf{a}_x^{ref})$. The CTC term promotes the target phonetic sequence, while the adversarial term λ_{GAN} encourages the modified representation $\hat{\mathbf{a}}_x$ to remain within the distribution of healthy speech plausible articulatory trajectories (resp. latent representation). Its associated discriminator D_ϕ contains three one-dimensional convolutional

²The code and model checkpoints can be found at <https://gricad-gitlab.univ-grenoble-alpes.fr/huebert/artimap>

³<https://github.com/Berkeley-Speech-Group/Speech-Articulatory-Coding>

⁴<https://hf.co/facebook/wav2vec2-lv-60-espeak-cv-ft>

¹Dysarthria can also affect prosody and phonation, speech rate, stress patterns, and phoneme voicing. Here, we focus on articulatory configuration.

layers with 256 hidden channels, uses a temporal context of two frames, and is trained with label smoothing of 0.1. Although the phonetic decoder and HiFi-GAN decoder remain frozen, gradients are propagated through both models to optimize the mapper.

4. EXPERIMENTS

4.1. Training data

Controlled synthetic speech ($\mathcal{D}_{\text{syn}}$): To pretrain the mapper, we generate parallel clean and artificial dysarthric speech from single-speaker TTS utterances⁵. Each clean utterance is analyzed using the frozen articulatory encoder of SAB to obtain a 12-dimensional articulatory trajectory, which is then selectively perturbed to simulate controlled articulatory deviations. To define these perturbations in an interpretable articulatory space, the trajectories are converted into six articulatory features following the procedure described in [9], which sequentially removes jaw contributions from the tongue and lip trajectories. For each articulatory dimension d at time t , trajectory excursions were modified around the corpus mean according to: $\tilde{\mathbf{a}}_t^{(d)} = \mu_d + \alpha_d(\mathbf{a}_t^{(d)} - \mu_d)$, where $\alpha_d = 1$ preserves the original trajectory and $0 < \alpha_d < 1$ compresses its excursion, producing hypo-articulated speech. The scaling factor α_d was either applied to all six articulatory features simultaneously or restricted to specific articulatory subsystems. We used $\alpha_d = 0.6$ for moderate perturbations and $\alpha_d = 0.3$ for severe perturbations on the selected articulatory dimensions. The resulting trajectories were converted back to speech using the remaining components of the SAB analysis-synthesis pipeline. We generated 5000 clean sentences ($\mathcal{D}_{\text{syn}}^{\text{(clean)}}$) and 10 000 perturbed sentences ($\mathcal{D}_{\text{syn}}^{\text{(dys)}}$) leading to a total of 7.46 h of speech. The synthetic test set comprises 1000 utterances and 28 298 reference phonemes, with 500 utterances per perturbation condition (moderate and severe).

Natural dysarthric speech ($\mathcal{D}_{\text{nat}}$): We use six dysarthric speakers from UASpeech [27]: M05, F04, M09, M08, M10, and F05, whose reported intelligibility scores range from 58 % to 95 %, and with a phoneme deletion rate below 0.20, thereby excluding speakers for whom phoneme omissions are frequent. This selection reflects the scope of our approach: the proposed mapper is designed to correct the articulatory realization of existing phonetic content, rather than recover phonemes that are absent from the input speech. After trimming the data with voice activity detection and manual inspection, we obtained a dataset of 4424 single-word recordings. To obtain longer inputs while retaining control over their lexical content, we construct three-word dysarthric sequences by concatenating isolated-word recordings from the same speaker. We then constructed training, validation and test set following a 70%, 15%, 15% split while ensuring that no lexical item occurs in more than one split. The resulting test set comprises 224 three-word sequences from the six selected speakers, containing 2897 reference phonemes.

4.2. Experimental setup

SAB and UAB mappers were both pretrained (stage 1) on the $\mathcal{D}_{\text{syn}}$ corpus (models SAB^{PT} and UAB^{PT}). Then, they were fine-tuned (stage 2) either on $\mathcal{D}_{\text{syn}}$ (models $\text{SAB}_{\text{syn}}^{\text{FT}}$ and $\text{UAB}_{\text{syn}}^{\text{FT}}$) or $\mathcal{D}_{\text{nat}}$ (models $\text{SAB}_{\text{nat}}^{\text{FT}}$ and $\text{UAB}_{\text{nat}}^{\text{FT}}$) corpus. We also include for evaluation two models whose mappers were trained with stage 2 only on $\mathcal{D}_{\text{nat}}$ (UAB_{nat} and SAB_{nat}), i.e., without pretraining on synthetic speech.

At evaluation time, we distinguish between the *identity* condition, obtained by analysis-resynthesis without applying the mapper, and the *enhanced* condition, obtained after applying the learned residual transformation. Models trained on $\mathcal{D}_{\text{syn}}$ are evaluated on both $\mathcal{D}_{\text{syn}}$ and $\mathcal{D}_{\text{nat}}$ test sets, with the latter considered zero-shot evaluation. Models fine-tuned on $\mathcal{D}_{\text{nat}}$ are evaluated only on this test set.

Training identity UAB (stage 0): The UAB *identity* projection and decoder are jointly pretrained on LibriTTS-R for 1.5M steps, using an initial 10k-step warm-up with an ℓ_1 log-mel reconstruction loss, followed by the HiFi-GAN objective combining mel reconstruction, adversarial losses, and feature matching, with respective weights of 45, 1, and 2. The model is trained using AdamW with a batch size of 16 and one-second waveform segments.

Pre-training SAB and UAB mappers (stage 1): Each mapper minimizes MSE over valid frames between mapped pathological codes and frame-aligned healthy targets from $\mathcal{D}_{\text{syn}}^{\text{(dys)}}$ and $\mathcal{D}_{\text{syn}}^{\text{(clean)}}$, respectively, extracted using the corresponding frozen identity encoder and projection. Training uses Adam (batch size 32), a cosine learning-rate schedule from 10^{-3} to 10^{-5} , and gradient clipping at norm 5. We train for up to 50 epochs with early stopping after 10 epochs without validation MSE improvement.

Fine-tuning SAB and UAB mappers (stage 2): SAB and UAB mappers are trained using Adam with a learning rate of 10^{-4} and a batch size of 4. Their discriminators are optimized separately with a learning rate of 10^{-4} . Fine-tuning runs for up to 120 epochs and validation CTC loss is evaluated every 2 epochs and training stops after 10 evaluations without improvement greater than 0.002. The checkpoint with the lowest validation loss is selected. The adversarial term is weighted by $\lambda_{\text{GAN}} = 0.05$.

4.3. Evaluation metrics

Automatic metrics: Phonetic intelligibility is assessed using phoneme error rate (PER). Phoneme sequences are predicted with the same phoneme recognizer as in stage 2 training. We additionally report corpus-level word error rate (WER), transcribing with Whisper large-v3⁶ using greedy decoding (English, transcription task, no timestamps). References and hypotheses are normalized using the Whisper English text normalizer.

Perceptual evaluation: We conducted an online intelligibility experiment with 48 native English-speaking participants recruited through Prolific. The experiment comprised three sections. First, participants transcribed 10 synthetic sentences (five moderate and five severe) presented in four conditions: Healthy TTS, synthetic dysarthric speech, and enhancement with the unsupervised and speech-articulatory bottlenecks ($\text{UAB}_{\text{syn}}^{\text{FT}}$ and $\text{SAB}_{\text{syn}}^{\text{FT}}$). Condition assignments for the synthetic sentences followed a Latin-square rotation across participants, while sentence order was randomized. Second, participants transcribed 15 isolated words from natural pathological speech, each presented in one of three conditions: original, UAB enhancement ($\text{UAB}_{\text{nat}}^{\text{FT}}$), or SAB enhancement ($\text{SAB}_{\text{nat}}^{\text{FT}}$). Condition assignment for these words was likewise rotated across participants. Finally, the same 15 words were presented in all three conditions with their corresponding target text, with their order randomized, and participants ranked the recordings by intelligibility. Transcription intelligibility was measured using WER. Participants were excluded if their WER on the intelligible TTS controls exceeded 25%. Audio examples are available on the accompanying demonstration page.⁷

⁵https://hf.co/espnets/kan-bayashi_ljspeech_joint_train-conformer_fastspeech2_hifigan

⁶<https://hf.co/openai/whisper-large-v3>

⁷<https://www.ultraspeech.com/artimap/demo.html>

Table 1. Corpus-level PER and Whisper WER (%) for SAB and UAB on $\mathcal{D}_{\text{syn}}$ and $\mathcal{D}_{\text{nat}}$ test sets. Identity denotes resynthesis without the mapper; other model rows report enhanced outputs. $\pm$ denotes the half-width of the 95% CI; $^{\text{ZS}}$ denotes zero-shot evaluation.

Model	$\mathcal{D}_{\text{syn}}$ test set		$\mathcal{D}_{\text{nat}}$ test set	
	PER	WER	PER	WER
Healthy	6.9 $\pm$ 0.4	4.8 $\pm$ 1.0	–	–
Dysarthric	32.0 $\pm$ 1.4	35.5 $\pm$ 3.0	31.3 $\pm$ 2.8	44.1 $\pm$ 18.6
UAB Identity	35.0 $\pm$ 1.4	38.5 $\pm$ 2.3	39.5 $\pm$ 3.5	53.4 $\pm$ 7.3
UAB ^{PT}	12.0 $\pm$ 0.6	10.6 $\pm$ 1.0	77.8 $\pm$ 4.5 ^{ZS}	125.0 $\pm$ 9.6 ^{ZS}
UAB ^{FT} _{syn}	9.4 $\pm$ 0.6	9.8 $\pm$ 1.0	73.3 $\pm$ 4.2 ^{ZS}	125.4 $\pm$ 10.0 ^{ZS}
UAB ^{FT} _{nat}	–	–	34.5 $\pm$ 3.1	49.2 $\pm$ 5.9
UAB _{nat}	–	–	36.9 $\pm$ 3.1	51.0 $\pm$ 6.3
SAB Identity	35.1 $\pm$ 1.4	41.9 $\pm$ 2.2	50.1 $\pm$ 3.8	76.8 $\pm$ 7.7
SAB ^{PT}	20.8 $\pm$ 1.0	23.0 $\pm$ 1.7	61.9 $\pm$ 4.6 ^{ZS}	133.1 $\pm$ 41.2 ^{ZS}
SAB ^{FT} _{syn}	17.4 $\pm$ 0.9	19.4 $\pm$ 1.5	71.0 $\pm$ 5.1 ^{ZS}	125.1 $\pm$ 9.4 ^{ZS}
SAB ^{FT} _{nat}	–	–	38.5 $\pm$ 3.1	78.9 $\pm$ 18.2
SAB _{nat}	–	–	42.6 $\pm$ 3.3	70.2 $\pm$ 7.7

5. RESULTS

Bottleneck mapping corrects synthetic perturbations. As shown in Table 1, MSE pretraining substantially reduces recognition errors. Compared with a $\mathcal{D}_{\text{syn}}$ PER of 32.0 % for the perturbed input, UAB^{PT} and SAB^{PT} achieve 12.0 % and 20.8 %, respectively. Fine-tuning with the combined CTC and adversarial objective brings further improvements, reaching 9.4 % for UAB^{FT}_{syn} and 17.4 % for SAB^{FT}_{syn}. A similar pattern is observed for WER, which decreases from 35.5 % for the input to 10.6 % and 23.0 % after pretraining, and to 9.8 % and 19.4 % after fine-tuning. The perceptual transcription experiment confirms the benefit of enhancement on synthetic speech: human WER decreased from 29.7 % for the synthetic dysarthric condition to 21.6 % with UAB^{FT}_{syn}. In contrast, SAB^{FT}_{syn} yielded a WER of 31.8 %, indicating that the objective improvements obtained with the articulatory mapper did not translate into a perceptual advantage in this setting. Clean TTS speech reached 5.0 % human WER, providing an upper-bound control for the perceptual evaluation.

Adaptation to natural speech enables correction within the resynthesis pipeline. Zero-shot (ZS) experiments involving models trained on synthetic only speech yield $\mathcal{D}_{\text{nat}}$ PER values between 61.9 % and 77.8 %, above their identity baselines, suggesting that controlled perturbations in $\mathcal{D}_{\text{syn}}^{(\text{dys})}$ do not fully capture natural dysarthric variability, and that adaptation to natural speech is necessary. After fine-tuning on $\mathcal{D}_{\text{nat}}$, PER decreases from 39.5 % to 34.5 % for UAB and from 50.1 % to 38.5 % for SAB relative to their identity conditions. The improvements after natural-speech fine-tuning show that the mapper can learn from target-domain data. Comparison between UAB^{FT}_{nat} and UAB_{nat} (resp. SAB^{FT}_{nat} and SAB_{nat}) shows the benefit on PER of model pretraining on synthetic speech. However, all enhanced systems remain above the original-input PER of 31.3 %. Word-level results also differ: WER decreases from 53.4 % to 49.2 % for UAB, while increasing from 76.8 % to 78.9 % for SAB, compared with 44.1 % for the original recordings. Surprisingly, the stage 1 pretraining is not beneficial in this case. The human transcription experiment showed the same overall tendency on natural pathological speech: human WER was 53.3 % for the original recordings, 55.8 % after UAB enhancement, and 62.9 % after SAB enhancement. In terms of intelligibility ranking, Wilcoxon

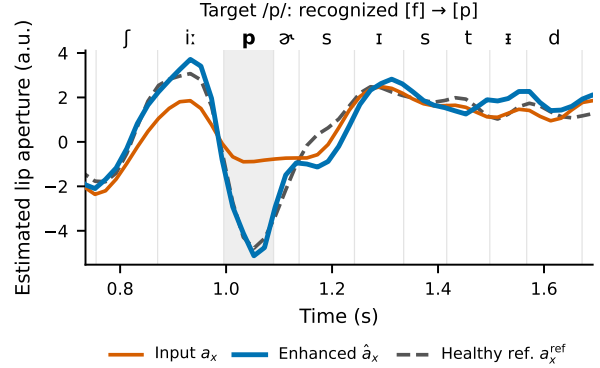


Fig. 2. Estimated lip aperture for a synthetic test utterance, “She persisted.” Around the target /p/ (shaded), the trajectory reconstructed by SAB^{FT}_{syn} approaches the healthy reference, accompanying a change in recognition from [f] to [p].

tests showed that the original recordings received significantly better ranks than SAB ($p = .0026$). No significant differences were observed between the original recordings and UAB, or between UAB and SAB ($p = .074$ for both comparisons).

Both bottlenecks support correction, while SAB allows changes to be interpreted in articulatory terms. The comparison between SAB and UAB provides insight into the role of explicit articulatory supervision. On synthetic speech, their identity PER values are nearly identical (35.1 % and 35.0 %), whereas UAB achieves lower recognition error after both mapper training stages. SAB nevertheless offers an interpretable structure: its dimensions correspond to estimated articulator trajectories, allowing the mapper’s changes to be inspected. This provides a basis for investigating how bottleneck modifications relate to phonetic correction. Fig. 2 displays an example of the lip aperture extracted from dysarthric ($\mathbf{a}_x$), reconstructed ($\hat{\mathbf{a}}_x$) and healthy reference ($\mathbf{a}_x^{\text{ref}}$) AB with the SAB^{FT}_{syn} app. Around the target /p/ (shaded), the trajectory reconstructed by SAB^{FT}_{syn} approaches the healthy reference.

6. CONCLUSION

This work introduces the first approach to explicitly modify articulatory representations for dysarthric speech reconstruction within a frozen neural analysis–synthesis framework. We address here the root cause of reduced intelligibility in dysarthria, rather than merely reconstructing linguistic content. We demonstrate an effective curriculum training pipeline: pretraining the mapper with MSE on synthetic speech provides an efficient initialization while fine-tuning with a phoneme-guided CTC objective on dysarthric speech then enables adaptation without requiring parallel healthy references. This pipeline achieves strong performance on synthetic data, validating the feasibility of corrective transformations within a compact bottleneck. While transferring these improvements to natural dysarthric speech remains challenging, we attribute this primarily to the scarcity of high-quality dysarthric data. Crucially, our method introduces an interpretable bottleneck, where each dimension corresponds to a specific articulatory trajectory. We believe that pursuing this direction with newly recorded data will enable precise, localized modifications with significant applications in speech rehabilitation, including visualization and coaching tools for clinical practice.

7. ACKNOWLEDGMENT

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8. ETHICAL COMPLIANCE AND AI USE DISCLOSURE

Speech corpora were used under their respective terms of use. Prolific participants provided informed consent and received compensation. Tools from Anthropic, OpenAI, and Mistral assisted with manuscript editing and code development. The authors reviewed and edited the resulting content and take full responsibility for the publication.

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